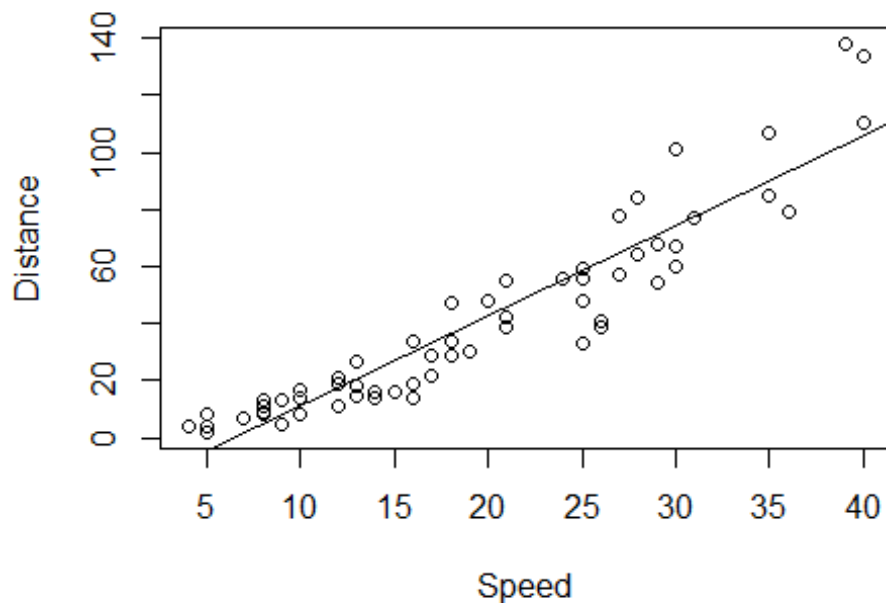


## Assignment-10.R

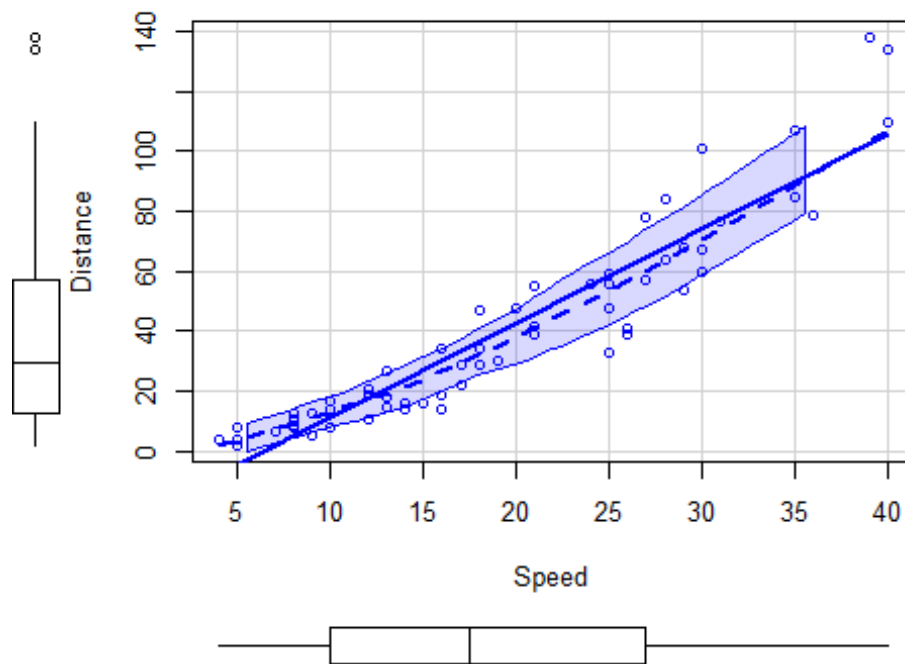
Ganga(54875)

2022-11-28

```
# Assignment-10 #  
# 7.6.1  
library(alr4)  
  
data(stopping)  
# visualizing the given 'stopping' data  
# the data consists of two variables:- distance as response and speed as  
# predictor  
plot(Distance~Speed,stopping)  
abline(lm(Distance~Speed,stopping))
```

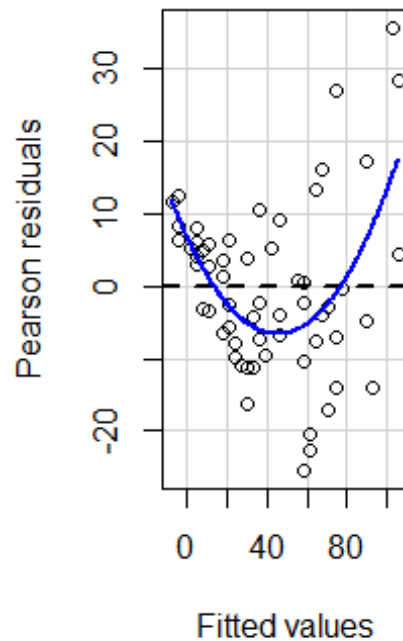
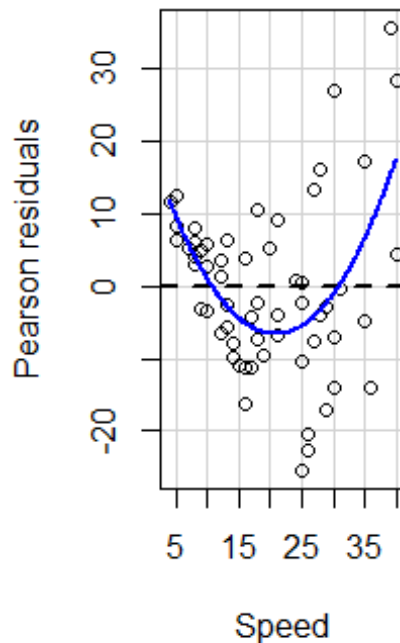


```
# according to the plot, the plot is linear and the variance is not constant  
# which implies that when the speed is less then the observations are near to  
# line with less variance and if speed is high then there is more variance  
# so, our assumption is wrong as a linear model.by analyzing the plot i see  
# that the distribution of data is in 'U' curvy path which shows there maybe  
# quadratic path.  
scatterplot(Distance~Speed,stopping)
```



# the dashed line shows that our assumption of quadratic path is correct.so,  
# Lets fit function using quadratic method.

```
X<-lm(Distance~Speed, stopping)
residualPlots(X)
```



```
##          Test stat Pr(>|Test stat|)
## Speed          5.0328      4.835e-06 ***
## Tukey test      5.0328      4.834e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

*# Residual analysis also suggests that the given model has the p-value nearly zero. so, we reject null hypothesis and the plot shows quadratic function.*

*#lets fit function using quadratic method.*

*# 7.6.2*

```
m0<-lm(Distance~Speed+I(Speed^2),stopping)
```

*# Lets do the variance test*

*#(A) Variance depending on the mean.*

```
ncvTest(m0)
```

```
## Non-constant Variance Score Test
```

```
## Variance formula: ~ fitted.values
```

```
## Chisquare = 22.97013, Df = 1, p = 1.6454e-06
```

*# According to the non constant variance test for the fitted values of the given model has highest value of test statistic(22.97013) and p-value is less than*

*# 0.05 of alpha which is  $p = 1.6454e-06$ (closer to zero). so, we reject the null hypothesis which means that the variance in the direction of mean is not constant.*

```

#(B) testing variance depends on Speed.
ncvTest(m0,~Speed)

## Non-constant Variance Score Test
## Variance formula: ~ Speed
## Chisquare = 23.39216, Df = 1, p = 1.3212e-06

# same as before the test statistic is 23.4 and p value is approx.zero. we
# reject the null hypothesis which means that the variance in the direction
# of variable speed is not constant.

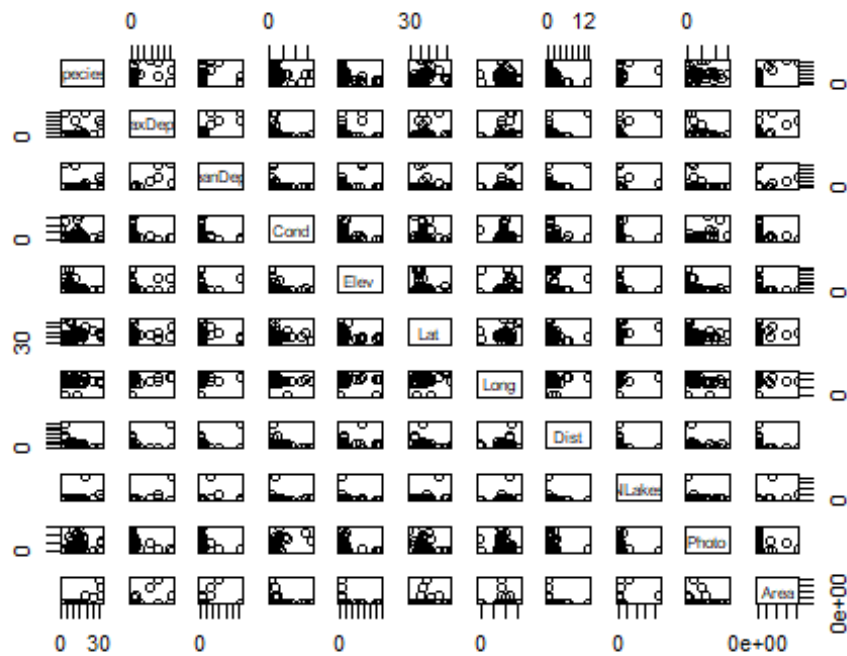
#(C) Test on variance depending on Speed and Speed2.
ncvTest(m0,~Speed+I(Speed^2))

## Non-constant Variance Score Test
## Variance formula: ~ Speed + I(Speed^2)
## Chisquare = 23.46559, Df = 2, p = 8.0262e-06

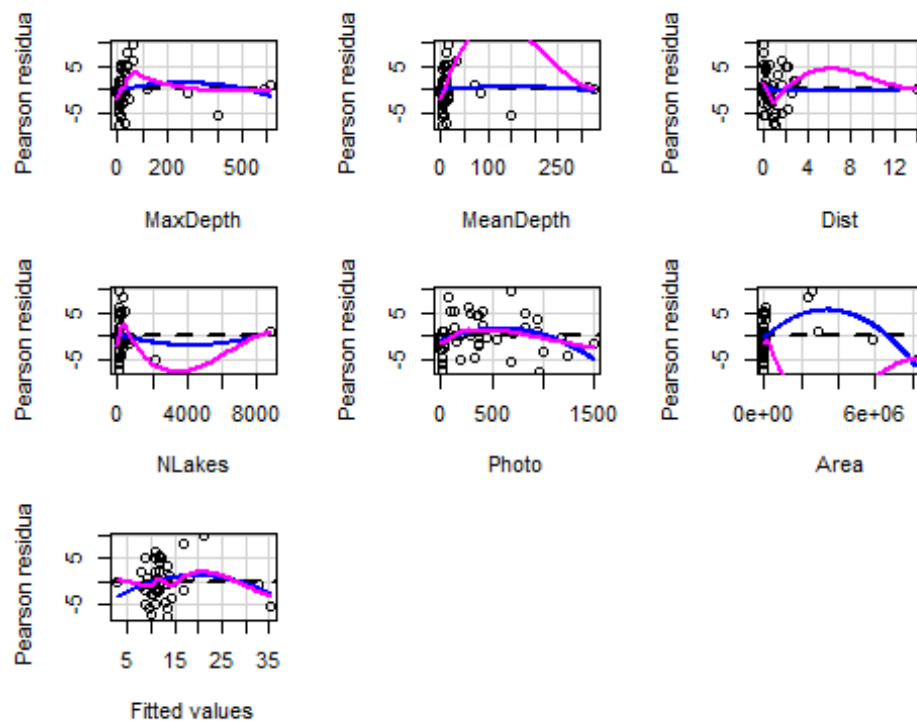
# The test of non-constant variance shows the test statistic=23.46559, p-
value
# as approx zero less than quantile alpha value(0.05). so, here we reject the
# null hypothesis as before which means that the variance in the direction of
the
# quadratic function is also not constant. so, even though we see a quadratic
# path in the distribution of the scatterplot its not having the constant
variance
# which is bad for the prediction("Adding quadratic term is not helpful").

# Problem -9.9
data(lakes)
View(lakes)
# lets visualize the data
pairs(Species~.,lakes)

```



```
# According the visualization i saw more pattern in most of the plots
# but when comparing with species i see a linear pattern with
Dist,NLakes,Photo
# and Area but there are some limitations like Leverage points in Dist plots
etc.
# note: I think its better removing the predictors Latitude, Longitude and
cond because we cannot transform them.
# Lets confirm the model is the better fit or not
Y<-lm(Species~MaxDepth+MeanDepth+Dist+NLakes+Photo+Area,lakes)
residualPlots(Y,smooth=T)
```



```
##          Test stat Pr(>|Test stat|)
## MaxDepth    -2.0575      0.0463642 *
## MeanDepth   -0.6982      0.4891854
## Dist         0.1943      0.8469515
## NLakes       0.8552      0.3976678
## Photo       -2.1903      0.0345455 *
## Area        -3.3884      0.0016191 **
## Tukey test   -3.4069      0.0006571 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

*# by analyzing the residual plots we see that for the tukey test and area plots*

*# the quadratic line is not near zero we see a pattern.the test statistic values are high*

*#(-3.3,-3.4) and the p-value is nearly zero less than alpha value. so we reject the null hypothesis.*

*# which results in Powertransformation.*

```
pt0<-
```

```
powerTransform(cbind(lakes$Species,lakes$MaxDepth,lakes$MeanDepth,lakes$Dist,
lakes$NLakes,lakes$Photo,lakes$Area))
summary(pt0)
```

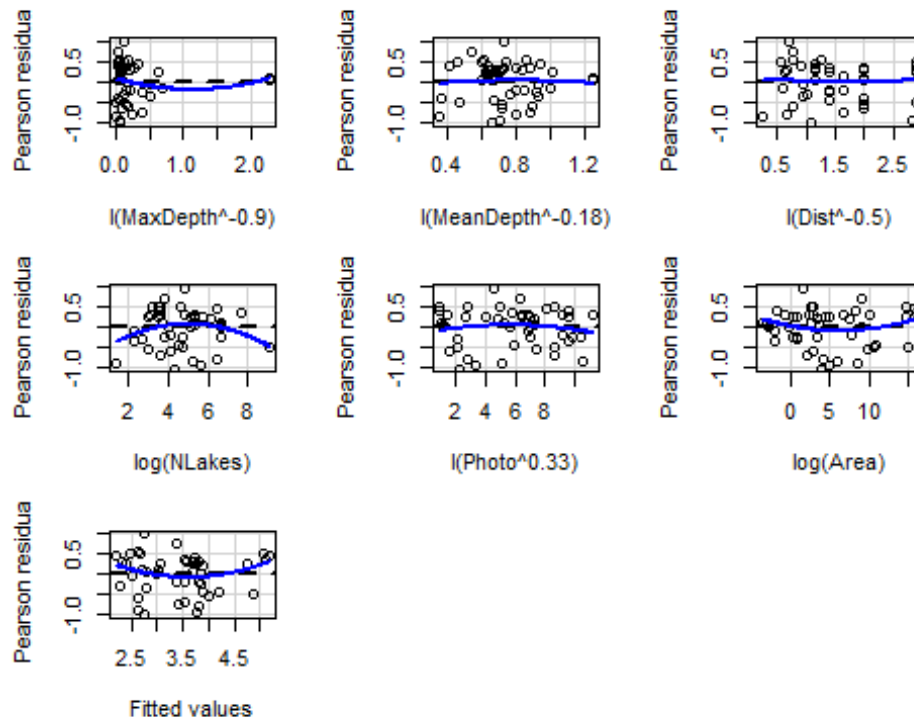
```
## bcPower Transformations to Multinormality
```

```
##      Est Power Rounded Pwr Wald Lwr Bnd Wald Upr Bnd
## Y1      0.4241          0.50      0.0712      0.7770
```

```
## Y2    -0.0921      -0.09      -0.1775      -0.0066
## Y3    -0.1793      -0.18      -0.2731      -0.0855
## Y4    -0.2706      -0.50      -0.5362      -0.0051
## Y5    -0.1402       0.00      -0.2870       0.0065
## Y6     0.2053       0.33       0.0601       0.3505
## Y7    -0.0469       0.00      -0.0995       0.0056
##
## Likelihood ratio test that transformation parameters are equal to 0
## (all log transformations)
##                               LRT df      pval
## LR test, lambda = (0 0 0 0 0 0 0) 47.21759  7 5.0622e-08
##
## Likelihood ratio test that no transformations are needed
##                               LRT df      pval
## LR test, lambda = (1 1 1 1 1 1 1) 1534.373  7 < 2.22e-16

# the new transformed values are (0.50 , -0.9, -0.18, -0.50, 0.00, 0.33, 0.00).
# where we get two
# predictor transformation as logs. according to the hypothesis tests we see
# that
# we reject null hypothesis for both lambda as (0000) and (1111) using p
# value < alpha.

# lets build the model using the new transformations
m1 <- lm(I(Species^0.50) ~ I(MaxDepth^-0.9) + I(MeanDepth^-0.18) + I(Dist^-
0.50) + log(NLakes) + I(Photo^0.33) + log(Area), lakes)
residualPlots(m1)
```

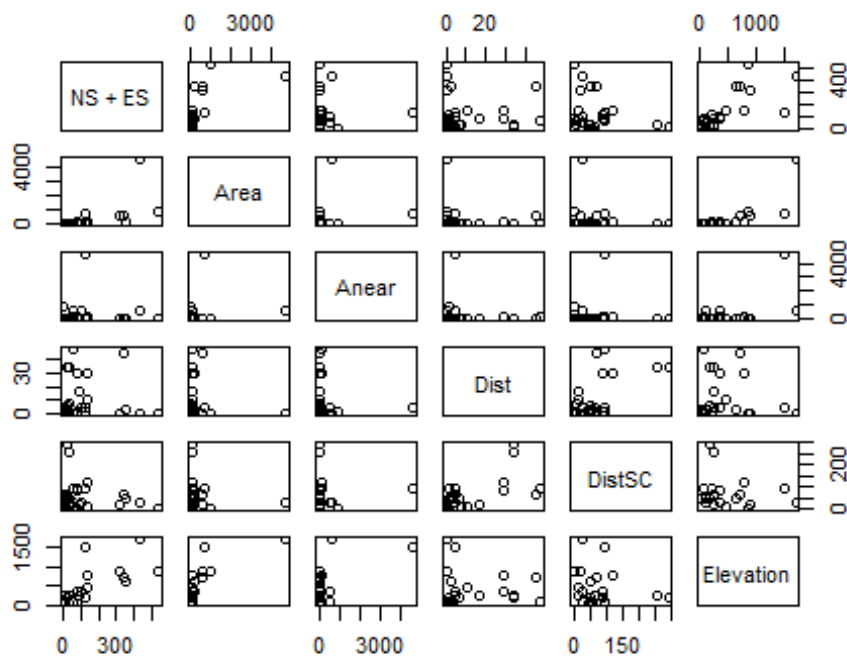


```
##              Test stat Pr(>|Test stat|)
## I(MaxDepth^-0.9)      1.2470      0.21985
## I(MeanDepth^-0.18)    -0.8960      0.37576
## I(Dist^-0.5)          0.2890      0.77412
## log(NLakes)           -1.5857      0.12089
## I(Photo^0.33)         -0.7968      0.43039
## log(Area)             1.6530      0.10636
## Tukey test            1.8331      0.06679 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# According to the residual analysis we see the t value is less than quantile
# value
# and p value is more than the alpha value which shows the variables for the
# residual
# test is not significant. i see a small significance but its not much
# harmful
# the new model is very good comparing to the old one, which means we do not
# reject
# the null hypothesis so all the plots has no pattern, but we are not 100%
# sure for prediction
# I can only say that this model is good compared to the previous ones.

# problem 10.6
# Given data is galapagos
data(galapagos)
# Visualizing the given data
pairs(NS+ES~Area+Anear+Dist+DistSC+Elevation,galapagos)
```





*# According to the plots i see a strong linear relation with elevation  
 # remaining all are also linear but they have some issues like  
 leverage,outliners  
 # also they are asymmetric.  
 # Lets summarize the results*

```
md=lm(NS+ES~Area+Anear+Dist+DistSC+Elevation,galapagos)
summary(md)
```

```
##
## Call:
## lm(formula = NS + ES ~ Area + Anear + Dist + DistSC + Elevation,
##     data = galapagos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -137.77  -46.49  -10.73   48.50  195.53
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  27.21286   35.24768   0.772  0.45069
## Area        -0.03543    0.03109  -1.140  0.27029
## Anear       -0.09568    0.02546  -3.759  0.00157 **
## Dist        -0.07577    1.42756  -0.053  0.95829
## DistSC      -0.37656    0.31383  -1.200  0.24664
## Elevation    0.40321    0.07978   5.054 9.78e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 81.55 on 17 degrees of freedom
## (6 observations deleted due to missingness)
## Multiple R-squared: 0.7819, Adjusted R-squared: 0.7178
## F-statistic: 12.19 on 5 and 17 DF, p-value: 3.951e-05

qt(1-0.05/2,17)

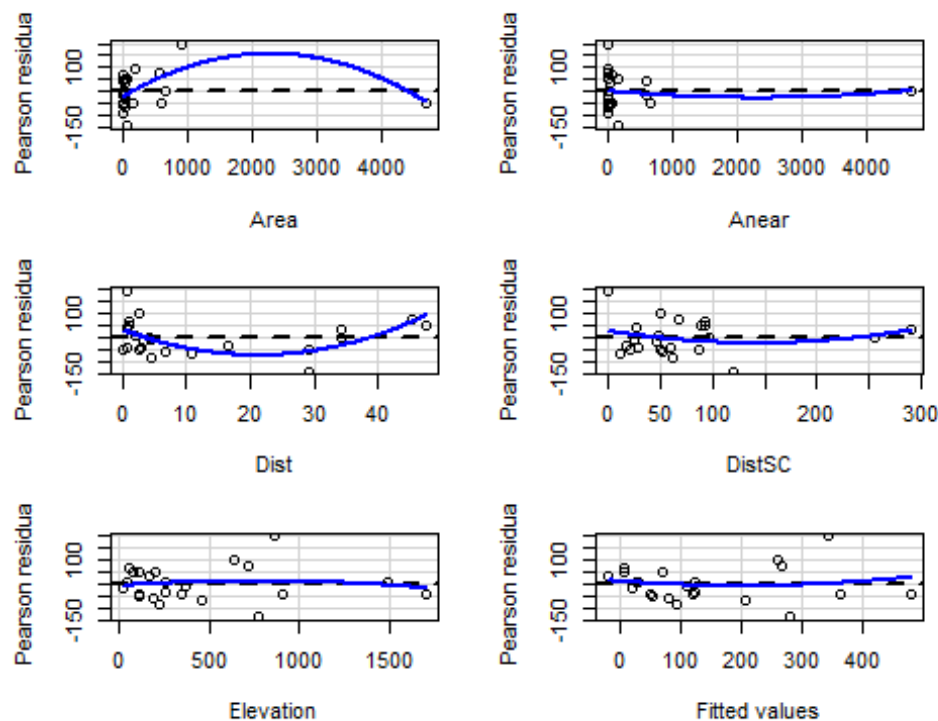
## [1] 2.109816

# According to the summary, the estimated T-value of the predictors Anear and
# Elevation has higher values(-3.759,5.054) than the significance level of
# quantile  $\alpha=0.05$ 
# which is 2.109816.Also the p-values of these regressors are 0.00157 and
# 9.78e-05 which is
# less than the significance quantile value of  $\alpha$ . so, these variables are
# significant
# means we reject null hypothesis. so these regressors have some effect on
# the diversity of
# the two species from the variables NS+ES.

# According to the question we have four possibilities
# (1) find the elevations;
# (2) delete these six islands from the data;
# (3) ignore elevation as a predictor of diversity,
# (4) substitute a plausible value for the missing data.

# my answer is selecting (2) and (3) is not a good idea because we know that
# elevation strongly effects the response and we know that these missing
# values
# are less than 200 which means that these values are not an outliers or
# Leverage
# points(influential points) it really effects the model.so i personally
# think prediction is the best thing i can do here so i choose (1).

residualPlots(md)
```



```
##          Test stat Pr(>|Test stat|)
## Area      -4.2540      0.0006061 ***
## Anear       0.2981      0.7695002
## Dist       2.9668      0.0090871 **
## DistSC      0.9279      0.3672466
## Elevation  -1.7213      0.1044654
## Tukey test  1.0453      0.2958988
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

*# for the prediction i used the residual analysis to check whether is it  
# a good model or not and the result shows that the area and dist has a  
# significant effect  
# in residual analysis and also the plots shows any pattern.so, its time to  
# transformation.*

*# using powertransformation*

```
pt2<-
```

```
powerTransform(cbind(galapagos$NS,galapagos$ES,galapagos$Area,galapagos$Anear,  
galapagos$Dist,galapagos$Elevation))
```

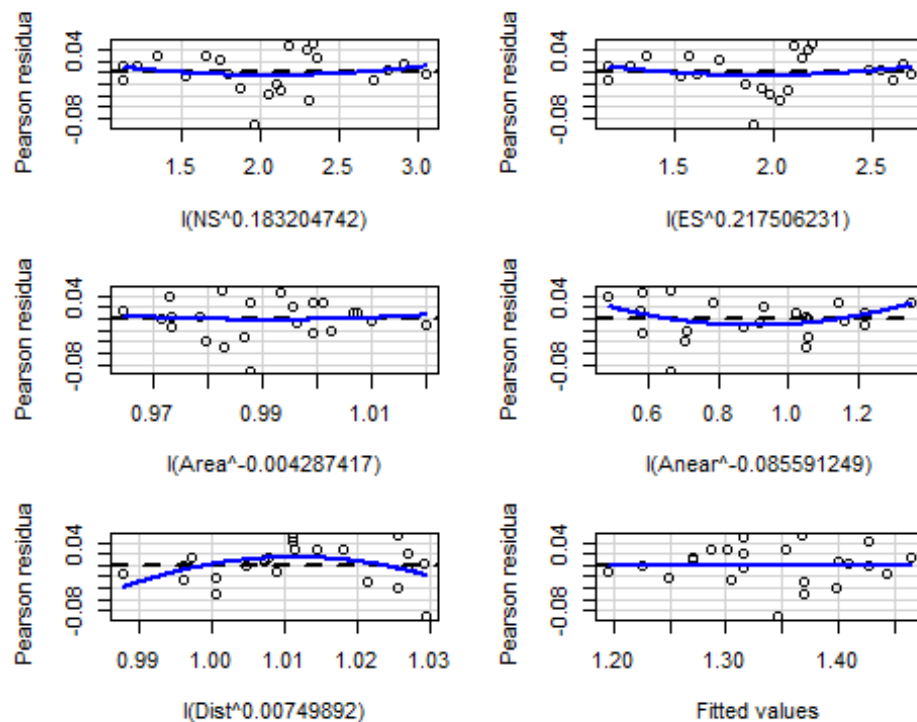
*# these are the results of my power transformation.*

```
#Y1          Y2          Y3          Y4          Y5          Y6  
#0.183204742  0.217506231 -0.004287417 -0.085591249  0.007498920  
0.052529105
```

*# lets build a model using the new values of our power transformation*

```
mt=lm(I(Elevation^0.052529105)~I(NS^0.183204742)+I(ES^0.217506231)+I(Area^-
```

```
0.004287417)+I(Anear^-0.085591249)+I(Dist^0.007498920),galapagos)
residualPlots(mt)
```



```
##               Test stat Pr(>|Test stat|)
## I(NS^0.183204742)      0.8648      0.39992
## I(ES^0.217506231)      0.9646      0.34908
## I(Area^-0.004287417)    0.2968      0.77040
## I(Anear^-0.085591249)   1.3284      0.20270
## I(Dist^0.00749892)     -2.3240      0.03362 *
## Tukey test              0.0441      0.96479
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# i genuinely tried the transformation for the variable distSC but i couldn't
transform it
# but according to the new residual analysis we see that the there is no
significance of
# quadratic term(for dist slight significance but not an issue). we get the
result using the p-values
# which are greater than the quantile value.so, we do not reject the null
hypothesis.
# According to the plots i don't see any pattern.so, its better model when
compared with
# the previous but im not 100% sure but its a better model.
```

```
# Lets do the prediction of the elevation
# FITTED AND SINGLE VALUES CONFIDENCE INTERVAL FOR BALTRA
```

```

predict(mt,
newdata=data.frame(NS=c(58),ES=c(23),Area=c(25.09),Anear=c(1.84),Dist=c(0.6))
,interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 1.367452 1.334566 1.400339

predict(mt,
newdata=data.frame(NS=c(58),ES=c(23),Area=c(25.09),Anear=c(1.84),Dist=c(0.6))
,interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 1.367452 1.280693 1.454211

# FITTED AND SINGLE VALUES CONFIDENCE INTERVAL FOR COAMANO
predict(mt,
newdata=data.frame(NS=c(2),ES=c(1),Area=c(0.05),Anear=c(903.82),Dist=c(1.9)),
,interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 1.197976 1.114103 1.281848

predict(mt,
newdata=data.frame(NS=c(2),ES=c(1),Area=c(0.05),Anear=c(903.82),Dist=c(1.9)),
,interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 1.197976 1.081872 1.31408

# FITTED AND SINGLE VALUES CONFIDENCE INTERVAL FOR Daphne_Major
predict(mt,
newdata=data.frame(NS=c(18),ES=c(11),Area=c(0.34),Anear=c(1.84),Dist=c(8.0)),
,interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 1.268801 1.234801 1.3028

predict(mt,
newdata=data.frame(NS=c(18),ES=c(11),Area=c(0.34),Anear=c(1.84),Dist=c(8.0)),
,interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 1.268801 1.181614 1.355987

# FITTED AND SINGLE VALUES CONFIDENCE INTERVAL FOR Eden
predict(mt,
newdata=data.frame(NS=c(8),ES=c(4),Area=c(0.03),Anear=c(17.95),Dist=c(0.4)),i
nterval="confidence", level=0.95)

##          fit          lwr          upr
## 1 1.197568 1.141016 1.25412

```

```

predict(mt,
newdata=data.frame(NS=c(8),ES=c(4),Area=c(0.03),Anear=c(17.95),Dist=c(0.4)),i
nterval="prediction", level=0.95)

##          fit          lwr          upr
## 1 1.197568 1.099366 1.29577

# FITTED AND SINGLE VALUES CONFIDENCE INTERVAL FOR Las_Plazas
predict(mt,
newdata=data.frame(NS=c(12),ES=c(9),Area=c(0.23),Anear=c(25.09),Dist=c(0.5)),
interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 1.26625 1.223624 1.308876

predict(mt,
newdata=data.frame(NS=c(12),ES=c(9),Area=c(0.23),Anear=c(25.09),Dist=c(0.5)),
interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 1.26625 1.175352 1.357148

# FITTED AND SINGLE VALUES CONFIDENCE INTERVAL FOR Seymour
predict(mt,
newdata=data.frame(NS=c(44),ES=c(16),Area=c(1.84),Anear=c(25.09),Dist=c(0.6))
,interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 1.291008 1.25226 1.329756

predict(mt,
newdata=data.frame(NS=c(44),ES=c(16),Area=c(1.84),Anear=c(25.09),Dist=c(0.6))
,interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 1.291008 1.201862 1.380154

FinaleE <- as.table(rbind(c(1.367452), c(1.197976),
c(1.268801),c(1.197568),c(1.26625),c( 1.291008)))
dimnames(FinaleE) <- list( c("baltra", "coamano",
"Daphne_Major", "Eden", "Las_Plazas", "Seymour"),
mt$Elevation^0.0525291)

FinaleE

##          [,1]
## baltra      1.367452
## coamano     1.197976
## Daphne_Major 1.268801
## Eden        1.197568
## Las_Plazas  1.266250
## Seymour     1.291008

```

# Thankinmg you#